ENVIRONMENT & HEALTH

Open Access

This article is licensed under CC-BY-NC-ND 4.0

pubs.acs.org/EnvHealth

Article

ABCC1 Deficiency Mediates Susceptibility to Cd-Induced Hearing Loss via the NOX2/ROS Axis

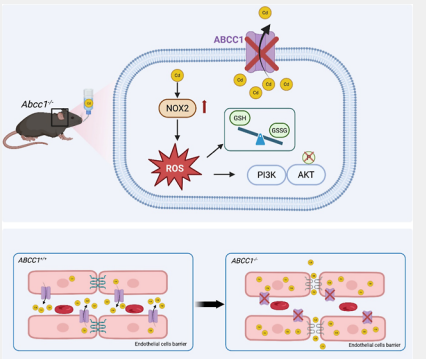
Shuai Zhang, Yongjia Chen, Yijiang Bai, Bo Pang, Mengzhu Jiang, Zequn Nie, Huping Huang, Lu Wang, Jie Ling, Tianhan Kai,*,¶ and Lingyun Mei*,¶

Cite This: https://doi.org/10.1021/envhealth.6c00120

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Cadmium (Cd), a prevalent environmental contaminant, is associated with auditory dysfunction. The cochlear blood−labyrinth barrier (BLB) protects the inner ear from toxic substances, a process in which the ATP-binding cassette subfamily C member 1 protein, ABCC1, plays a critical role in expelling toxins from its cells. To understand the specific contribution of ABCC1 to Cd-induced hearing damage, we employed both an in vivo model (ABCC1 knockout mice, *Abcc1*$^{-/-}$ mice) and an in vitro endothelial model (ABCC1-deficient human umbilical vein endothelial cells, *ABCC1*$^{-/-}$HUVECs). These models were exposed to $CdCl_2$ to assess auditory responses, histopathology, BLB integrity, and oxidative stress. We also explored the NOX2-ROS/PI3K-AKT pathway using immunofluorescence, Western blotting, and RNA-seq analyses. The results indicate that ABCC1 deficiency exacerbated Cd-induced BLB leakage, promoted spiral ganglion neurons and presynaptic ribbon puncta loss, and increased ABR thresholds. In both animal and cellular models, ABCC1 deficiency enhanced NOX2 expression, ROS accumulation, and AKT dephosphorylation. Significantly, pharmacological inhibition of NOX2 with GSK2795039 attenuated oxidative stress, preserved endothelial barrier function, and partially restored PI3K-AKT signaling. These findings identify ABCC1 as a critical protector of cochlear vascular integrity during Cd exposure, acting through the NOX2-ROS/PI3K-AKT axis, and suggest its potential as a therapeutic target for preventing Cd-induced hearing loss.



KEYWORDS: *Cd, ABCC1, blood−labyrinth barrier, NOX2, oxidative stress*

1. INTRODUCTION

Hearing loss affects approximately 1.5 billion people globally—about 20% of the world's population—a figure projected to rise to 2.5 billion by 2050.[1] Its etiology is complex, involving interactions between genetic predisposition and environmental factors.[2,3] Despite the pervasive threat to hearing from various ototoxic elements—including noise, pharmaceuticals, and chemicals—in our contemporary world, systematic research into environmental risk factors beyond gene−noise interactions remains scarce. In particular, the impact of gene−heavy metal interactions on auditory function has received insufficient attention.[4,5]

While heavy metals originate naturally in the Earth's crust, rapid industrial growth has drastically amplified their presence in water and soil, creating severe ecological and public health hazards through food chain biomagnification.[6] A key heavy metal pollutant, cadmium (Cd), found in tobacco products and released by fossil fuel combustion, also enters the body via diet.[7] Daily intake often approaches the tolerable limit of 70 μg, and due to its exceptionally long biological half-life (>26 years), Cd progressively accumulates in human tissues.[8,9] In the United States, an epidemiological study involving 3698 individuals demonstrated that even low-level Cd exposure is linked to an increased risk of hearing loss in the general population.[10] Nonetheless, the majority of mechanistic investigations conducted to date have employed in vitro models, such as cultured cochlear spiral ganglion neurons or the organ of Corti,[11−13] which do not accurately reflect real-world exposure pathways, thus limiting their practical applicability. It should be noted that Cd concentrations used in experimental models are typically higher than those encountered in human environmental exposure. In such settings, elevated exposure levels are often applied to induce measurable biological effects within a limited experimental time frame, thereby modeling cumulative tissue burden rather than acute physiological concentrations. Human studies, while valuable, have largely treated heavy metals as isolated factors, often overlooking critical gene−environment interactions.

Our research group previously identified a novel deafness gene, ATP-binding cassette subfamily C member 1 (ABCC1), in a five-generation family with a history of hereditary deafness

Received: February 25, 2026
Revised: June 23, 2026
Accepted: June 24, 2026

ACS Publications

© XXXX The Authors. Co-published by Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences, and American Chemical Society

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581560/envhealth.6c00120.pdf by guest on 10 October 2026

in Hunan Province, China, where 10 family members were affected. The missense mutation in ABCC1 was found to segregate with the hearing loss phenotype in the family, following an autosomal dominant inheritance pattern. Clinically, affected individuals initially presented with bilateral high-frequency hearing loss, which progressively developed into bilateral, full-frequency sensorineural hearing loss.[14] ABCC1 is a transmembrane transporter that plays a critical role in cellular detoxification by exporting toxic metabolites, organic anions, and exogenous compounds—including certain heavy metals—out of the cell.[15−17] Although most studies on ABCC1 have focused on multidrug resistance, the molecular mechanism underlying ABCC1-related hearing loss remains unclear.

Notably, Hunan, a major nonferrous metal production and rice cultivation region, faces persistent Cd pollution concerns.[18] Environmental monitoring studies have revealed that both soil and rice in Hunan exhibit elevated levels of Cd.[19] Furthermore, an analysis of hair samples from 1202 residents indicated significantly higher Cd levels in the hair of individuals from Changsha, a representative city in Hunan.[20] These observations prompted us to investigate a potential link between Cd exposure and hereditary hearing loss associated with the ABCC1 dysfunction.

ABCC1 exhibits a broad spectrum of transport substrates including the heavy metal Cd. Clinical observations have revealed a statistically significant correlation between urinary Cd levels and ABCC1 genetic variants, suggesting that ABCC1 may play a role in modulating Cd distribution.[21] More direct evidence emerges from cellular studies: ABCC1-deficient zebrafish embryos accumulated 2.3-fold higher Cd levels compared to controls,[22] while ABCC1 overexpression enhanced cellular Cd efflux and alleviated Cd-induced cytotoxicity.[23] Although these findings provide substantial evidence that Cd is a transport substrate of ABCC1, the specific role of ABCC1 in Cd-induced ototoxicity remains unexplored. ABCC1 is expressed in endothelial cells of the cochlear blood−labyrinth barrier (BLB), a structure critical for maintaining inner ear homeostasis by blocking the entry of toxic substances and facilitating efflux of metabolic waste.[24] However, whether ABCC1 confers otoprotection via its Cd efflux function at the BLB requires further investigation.

In this study, we aimed to elucidate the role of ABCC1 in Cd-induced ototoxicity with a particular focus on its endothelial functions in the BLB. ABCC1 knockout mouse models and endothelial cell models were established to investigate oxidative stress responses and barrier integrity following Cd exposure. By integrating in vivo and in vitro data with RNA sequencing analysis, we further examined the potential involvement of the NOX2-ROS/PI3K-AKT signaling pathway. This work provides novel mechanistic insights into ABCC1-mediated protection against Cd toxicity and suggests potential strategies for the prevention and treatment of Cd-induced hearing loss and endothelial dysfunction.

2. METHOD

2.1. Animal Housing and Exposure

All experiments involving mice were approved by the Institutional Animal Care and Use Committee (IACUC) of Central South University. The *Abcc1*$^{-/-}$ mice were purchased from Shanghai Model Organisms Center, Inc., with genotyping performed using the method outlined in our previous publication.[24] Mice were housed under specific pathogen-free (SPF) conditions (4−5 mice per cage) with a 12 h light/dark cycle, 40%−50% relative humidity, and a stable temperature of 25 ± 1 °C, with free access to food and water.

Two-month-old *Abcc1*$^{-/-}$, *Abcc1*$^{+/-}$, and *Abcc1*$^{+/+}$ mice were allocated to experimental conditions and were either provided with drinking water containing $CdCl_2$ (in concentrations of 100, 400, 800, and 1600 ppm) or with water devoid of the compound. All groups, which included both genders, received fresh water every 5−7 days, enabling observation of their activity and disposition. After two months of breeding, their ABR thresholds were assessed.

2.2. Auditory Brainstem Response (ABR)

To prepare for ABR testing, mice received anesthesia via a 0.3% pentobarbital sodium solution, administered at 0.1 mL per 10 g of body weight. Subsequently, the mice were placed in a soundproof chamber for ABR testing. During the testing, their body temperature was maintained at 37 °C with a heating pad throughout the procedure. For signal acquisition, an active electrode was affixed centrally on the scalp, a reference was placed behind the tested ear, and a ground electrode secured on the back. MF1Multi-Field Magnetic Speakers (Tucker-Davis Technologies Inc., Alachua, FL, USA) were positioned approximately 10 cm from the tested ear, to deliver auditory stimuli. These speakers underwent calibration using a sensor signal conditioner (model 480C02, PCB, NY, USA) and RZ6Multi I/O Processor (Tucker-Davis Technologies Inc., Alachua, FL, USA) before the experiment. A Tucker-Davis Technologies System workstation equipped with a SigGen32 software (Tucker-Davis Technologies Inc., Alachua, FL, USA) was used to measure the ABR thresholds for various stimuli, including Click and pure tones from 4 kHz to 32 kHz. For each frequency, the sound intensity progressively decreased from 90 dB SPL down to 10 dB SPL. The resulting electrical signals, amplified by a TDT Medusa4Z biological amplifier (with a 30−3000 Hz bandpass filter and a 50 Hz notch filter), were then processed using BioSigRZ software (Tucker-Davis Technologies). Ultimately, the ABR threshold was determined as the lowest sound pressure level capable of generating a reproducible and discernible waveform.

2.3. BLB Permeability Assays

A fluorescent tracer, FITC-dextran (Table S1), was dissolved in saline at 5 mg/mL and injected into mice via the tail vein (0.1 mL per 10 g body weight). The tracer was permitted to circulate for 15 min postinjection, after which the mice were euthanized. Their cochleae underwent careful harvesting and overnight fixation in 4% paraformaldehyde (PFA). The next day, after three PBS washes, the stria vascularis was precisely microdissected, mounted on glass slides, and counterstained with DAPI to visualize cell nuclei. Images of the stria vascularis were subsequently acquired using a Zeiss confocal microscope.

2.4. Whole Mount Preparation and Immunofluorescence Staining

Mice were euthanized and decapitated for cochlear extraction. The isolated cochleae were transferred in a Petri dish containing 10 mmol/L PBS, and a small hole was made at the apex. The cochleae were perfused with 4% PFA through the round and oval windows and then fixed overnight in 4% PFA at 4 °C. On the following day, the cochleae were decalcified in 10% ethylenediaminetetraacetic acid (EDTA) for 3 days. Using microdissection techniques, the basilar

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581560/envhealth.6c00120.pdf by guest on 10 October 2026

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX

membrane was carefully isolated and flattened. The samples were permeabilized and blocked for 1 h in a solution containing 2% Triton X-100 and 5% bovine serum albumin (BSA), followed by incubation with a primary antibody (Table S2) diluted in 0.1% Triton X-100 and 5% BSA at 4 °C overnight. On the following day, after removing the primary antibody, the tissue was washed three times with 10 mmol/L PBS and then incubated with the corresponding secondary antibody. For basilar membrane visualization, the tissue was incubated with Phalloidin Fluor 488 for 30 min. Nuclei were counterstained with DAPI for 5 min, followed by three PBS washes. Finally, the tissues were mounted in antifade mounting medium to preserve fluorescence and imaged using a Zeiss confocal microscope. Images were analyzed with ZEN software (Zeiss) and ImageJ Fiji.

2.5. Cochlear Frozen Section Preparation and Immunofluorescence

Fixed cochleae were decalcified in 10% EDTA for 1 week, with the solution changed every 2 days. The decalcified tissues were sectioned at 10 μm thickness using a Leica CM1950 cryostat. After drying, sections were permeabilized and blocked with 0.1% Triton X-100 and 5% BSA for 1 h at room temperature, followed by incubation with the primary antibody at 4 °C overnight. The next day, after three washes with PBS, the sections were incubated with appropriate secondary antibodies for 1 h at room temperature. Post-PBS wash, samples were mounted with an antifade reagent for fluorescence preservation. Images were captured using a Zeiss confocal microscope and processed with Zen software (Zeiss) and ImageJ Fiji software.

2.6. Cell Culture and sgRNA Transfection to Construct *ABCC1*$^{-/-}$ HUVECs

Human umbilical vein endothelial cells (HUVECs) were cultured in Dulbecco's Modified Eagle's Medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and endothelial cell growth supplement. These cells were sustained at 37 °C within a humidified 5% CO_2 environment and passaged for 5−6 generations prior to use.

The human ABCC1 gene-specific sgRNA was purchased from YouBio Biotechnology Co., Ltd. For lentiviral particle generation, HEK293T cells were initially plated in six-well plates, grown to 70−80% confluence, and then cotransfected with the pLenti-CRISPR v2-ABCC1 sgRNA plasmid, pMD2.G, and psPAX2 packaging plasmids. After 48 h, viral supernatants were collected, purified by filtration through a 0.45 μm membrane to exclude cellular debris, and then used for HUVEC transduction. Following 48 h post-transduction, cells underwent a 7 day selection period using puromycin (3 μg/mL), with medium replacements every 2 days to eliminate nontransduced cells. Puromycin-resistant HUVECs were then isolated via single-cell cloning by limiting dilution, with each cell receiving one cell to guarantee clonality. Once achieving sufficient density, clones were expanded and their ABCC1 knockout efficiency was verified by Western blot analysis.

2.7. Western Blot Assay

Cellular proteins were extracted by lysing cells on ice with RIPA buffer supplemented with phosphatase and protease inhibitors. Protein concentration was quantified using a BCA protein assay kit. For separation, 20 μg of protein per lane underwent 8% or 10% sodium dodecyl sulfate−polyacrylamide gel electrophoresis (SDS-PAGE) before being transferred onto a 0.45 μm equilibrated polyvinylidene fluoride (PVDF) membrane. The membrane was blocked for 1−2 h with 5% BSA at room temperature, followed by incubation with the primary antibody at 4 °C overnight. The next day, the membrane was washed three times for 10 min with Tris-buffered saline containing Tween-20 (TBST) and then incubated with the secondary antibody at room temperature for 1 h. Protein signals were detected using enhanced chemiluminescence and captured with an Azure 500 imaging system. ImageJ software was utilized for subsequent image analysis. All experiments were independently repeated three times.

2.8. Cell Viability Assay

Cell viability was assessed using the Cell Counting Kit-8 (CCK-8). Cells were seeded in 96-well plates and allowed to adhere prior to $CdCl_2$ treatment. Treatment involved either varying concentrations (0−100 μmol/L) for 24 h or a constant 30 μmol/L for 24, 48, and 72 h. Post-treatment, 100 μL of complete culture medium and 10 μL of CCK-8 solution were introduced, followed by a 1 h dark incubation at 37 °C. Absorbance at 450 nm was measured using a multimode microplate reader, with culture medium for background subtraction. Three independent experiments were performed to assess cell viability based on absorbance measurements at 450 nm.

2.9. Endothelial Cell Permeability Assay

Cells were seeded into the upper chamber of a 12-well Transwell plate fitted with 0.4 μm pore membranes and cultured until cell confluence was achieved. The culture medium in both the upper and lower chambers was replaced with a complete medium with or without 30 μmol/L $CdCl_2$, followed by 24 h of incubation. The transendothelial electrical resistance (TEER) was assessed using a Millicell ERS-2 V/ohm meter (Millipore, Billerica, MA) according to the manufacturer's specifications. Each well was measured three times, and the average value was calculated. The TEER value for each experimental condition was derived by subtracting the resistance value of cell-free blank wells and multiplying by the surface area of the upper chamber (1.12 cm^2), with final results expressed in $\Omega \cdot cm^2$ units.

For permeability assessment, after medium removal, 500 μL of FITC-dextran (1 mg/mL) was added to the upper chamber, while 600 μL of fresh medium was placed in the lower chamber. The system was incubated in a cell culture incubator for 1 h, after which the solution in the lower chamber was collected. Fluorescence intensity was measured at 492 nm excitation and 520 nm emission using a multimode microplate reader.

2.10. Immunofluorescence Staining of Cells

After removing the culture medium, cells were washed with ice-cold PBS and fixed with 4% PFA at room temperature for 15 min. The cells were then blocked with 5% BSA for 1 h at room temperature, followed by incubation with the primary antibody at 4 °C overnight. After three 10 min washes with PBS on a shaker, the cells were incubated with a secondary antibody for 1 h at room temperature. Following three additional washes, cell nuclei were counterstained with DAPI. Finally, cells were mounted with an antifade mounting medium to preserve fluorescence and imaged using a Zeiss confocal microscope.

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581560/envhealth.6c00120.pdf by guest on 10 October 2026

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX

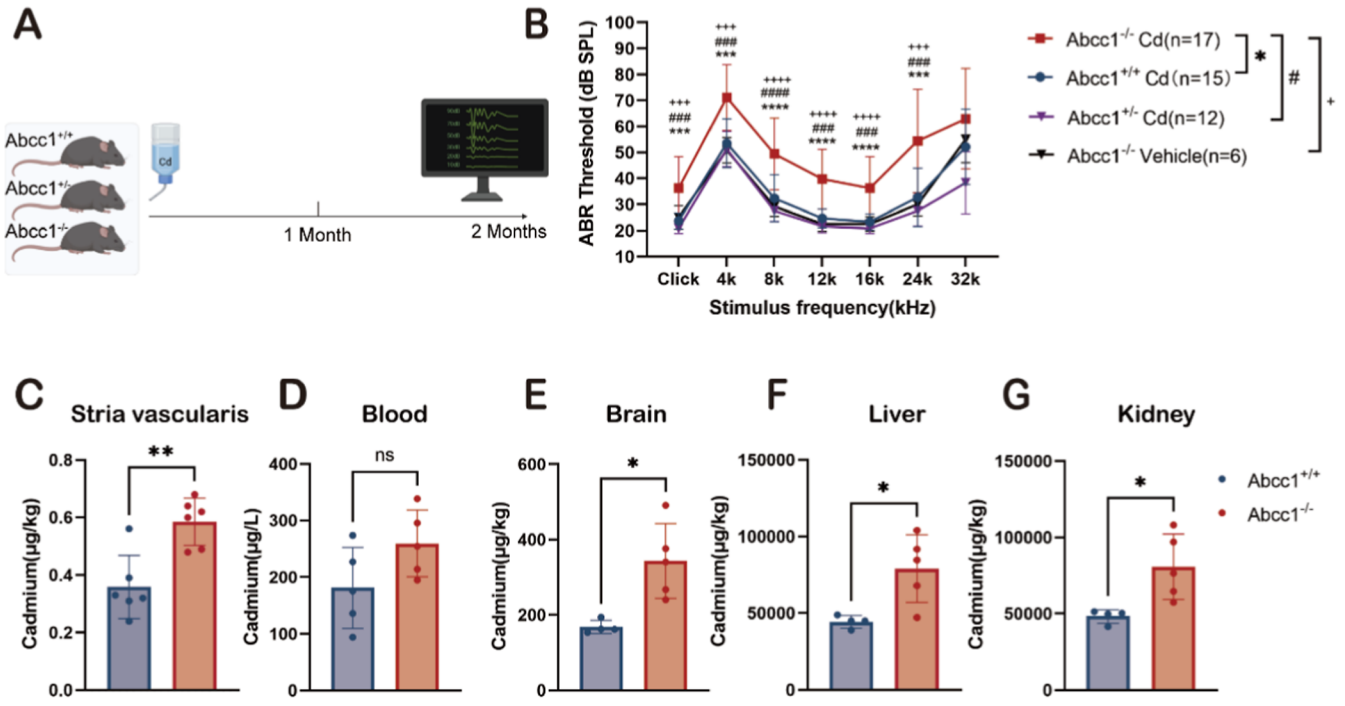


Figure 1. Auditory phenotypes and tissue Cd accumulation in mice. (A) Experimental workflow for mouse modeling. (B) Auditory brainstem response thresholds. Cd accumulation in the cochlear stria vascularis (C), circulating blood (D), brain tissue (E), liver tissue (F), and kidney tissue (G). The data are presented as mean ± SD; *p <0.05, **p <0.01, ***p <0.001, ****p <0.0001.

2.11. Transcriptome Sequencing and Analysis

Total RNA was extracted from *ABCC1*$^{+/+}$ and *ABCC1*$^{-/-}$ HUVECs following 24 h exposure to $CdCl_2$ using the TRIzol reagent for RNA sequencing (RNA-seq). Three independent biological replicates (n = 3) were generated from independently cultured cell samples.

For mouse cochlear stria vascularis, tissues were harvested from *Abcc1*$^{-/-}$ and *Abcc1*$^{+/+}$ mice after two months of continuous exposure to 800 ppm of $CdCl_2$ in drinking water. Three independent biological replicates were included per group (n = 3), and each replicate consisted of pooled stria vascularis tissues from 6 mice (12 cochleae) to ensure sufficient RNA quantity and reduce variability.

RNA quality was verified by assessing purity and quantification with a NanoDrop 2000 spectrophotometer (Thermo Scientific, USA) and integrity using an Agilent 2100 Bioanalyzer (Agilent Technologies, Santa Clara, CA, USA). Library preparation was performed using the VAHTS Universal V6 RNA-seq Library Prep Kit, and libraries were sequenced on an Illumina platform with an average depth of approximately 25 million reads per sample. Transcriptome sequencing and analysis were conducted by OE Biotech (Shanghai, China).

Differential expression analysis was performed using the DESeq2,[25] with p-values adjusted using the Benjamini–Hochberg method to control the false discovery rate (FDR). Genes with adjusted p-value <0.05 and |log2 fold change| >1 were considered significantly differentially expressed. Functional enrichment analyses (GO, KEGG, Reactome, and WikiPathways) were performed using R software, and Gene Set Enrichment Analysis (GSEA) was conducted using GSEA software.[26] The RNA-seq data generated in this study have been deposited in the NCBI Sequence Read Archive (SRA) under BioProject accession number PRJNA1423016.

2.12. RNA Extraction and qRT–PCR Analysis

Total RNA was extracted using the TRIzol reagent. Genomic DNA was removed, and first-strand cDNA was synthesized according to the manufacturer's instructions. Quantitative real-time PCR (qRT–PCR) was performed using a fluorescent probe-based kit. Relative gene expression levels were normalized to GAPDH. The primer sequences are shown in Table S3.

2.13. Statistical Analysis

One-way analysis of variance (ANOVA) was used to compare differences among three or more groups, followed by posthoc multiple comparisons using Tukey's test. Statistical significance was defined as a p-value of less than 0.05. All graphical representations were generated using GraphPad Prism 8, with data expressed as mean ± standard deviation (SD). Statistical analyses were performed using IBM SPSS Statistics 27.

3. RESULTS

3.1. $CdCl_2$ Exposure Induces Hearing Loss and Cochlear Cd Accumulation in *Abcc1*$^{-/-}$ Mice

Initial ABR testing on 5 month-old *Abcc1*$^{+/+}$, *Abcc1*$^{+/-}$, and *Abcc1*$^{-/-}$ mice under SPF conditions confirmed no significant ABR threshold variations across genotypes (Figure S1), indicating that any subsequent changes were due to environmental exposure rather than genetic background. On the basis of these results, we exposed 2 month-old mice of all genotypes to differing concentrations (100, 400, 800, and 1600 ppm) of $CdCl_2$ in their drinking water for 2 months (Figure 1A). Findings revealed that 1600 ppm of $CdCl_2$ caused mortality within one month, while 100 and 400 ppm showed no significant change in ABR thresholds over the 2 month period (Figures S2 and S3). Therefore, we used a concentration of 800 ppm of $CdCl_2$ in the following experiment to investigate

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX–XXX

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581560/envhealth.6c00120.pdf by guest on 10 October 2026

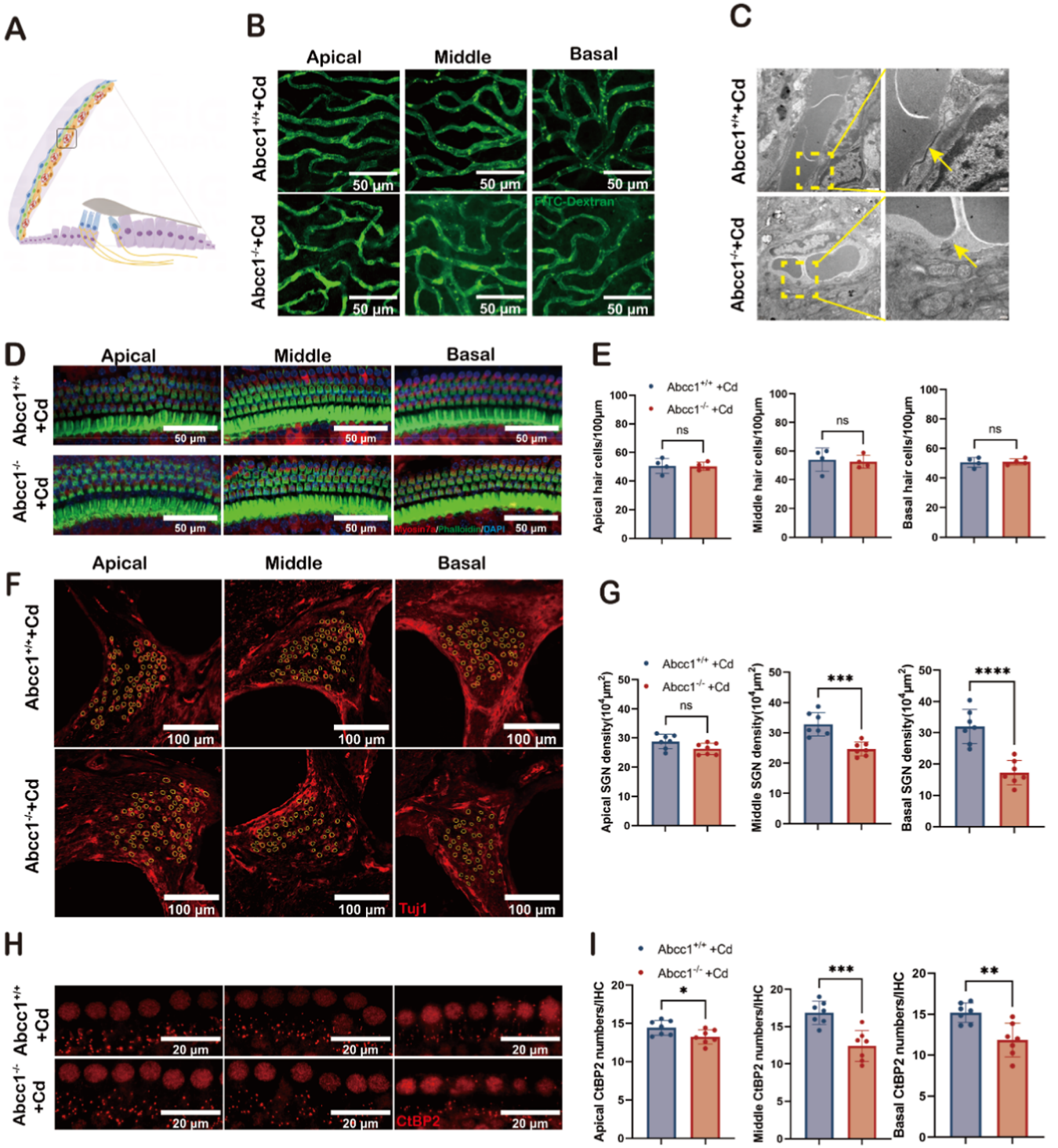


Figure 2. Morphological phenotypes of the mouse cochlea. (A) Schematic of a cochlear cross-section, with the boxed region indicating the stria vascularis. (B) Permeability staining of the stria vascularis. (C) Representative ultrastructure of the stria vascularis by electron microscopy. (D) Representative morphology of cochlear hair cells. (E) Quantification of hair cell numbers. (F) Representative staining of spiral ganglion neurons. (G) Quantification of spiral ganglion neuron numbers. (H) Representative staining of cochlear presynaptic ribbon puncta. (I) Quantification of presynaptic ribbon puncta numbers. Data are presented as mean ± SD; *p <0.05, **p <0.01, ***p <0.001, ****p <0.0001.

the role of the ABCC1 in Cd-induced ototoxicity. Comparison of ABR thresholds revealed that *Abcc1*$^{-/-}$ mice exhibited significantly higher threshold shifts from 4 kHz to 24 kHz than *Abcc1*$^{+/+}$ and *Abcc1*$^{+/-}$ mice (Figure 1B), which strongly suggests the protective role of ABCC1 against Cd-induced ototoxicity.

Analysis of Cd tissue distribution across different mouse genotypes revealed that the absence of ABCC1 significantly influences Cd accumulation. Specifically, *Abcc1*$^{-/-}$ mice exhibited 1.6-fold greater Cd levels in the stria vascularis compared to *Abcc1*$^{+/+}$ mice, suggesting that ABCC1 deficiency may promote Cd accumulation in the cochlea and increase susceptibility to hearing loss (Figure 1C), while no significant

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581156/envhealth.6c00120.pdf by guest on 10 October 2026

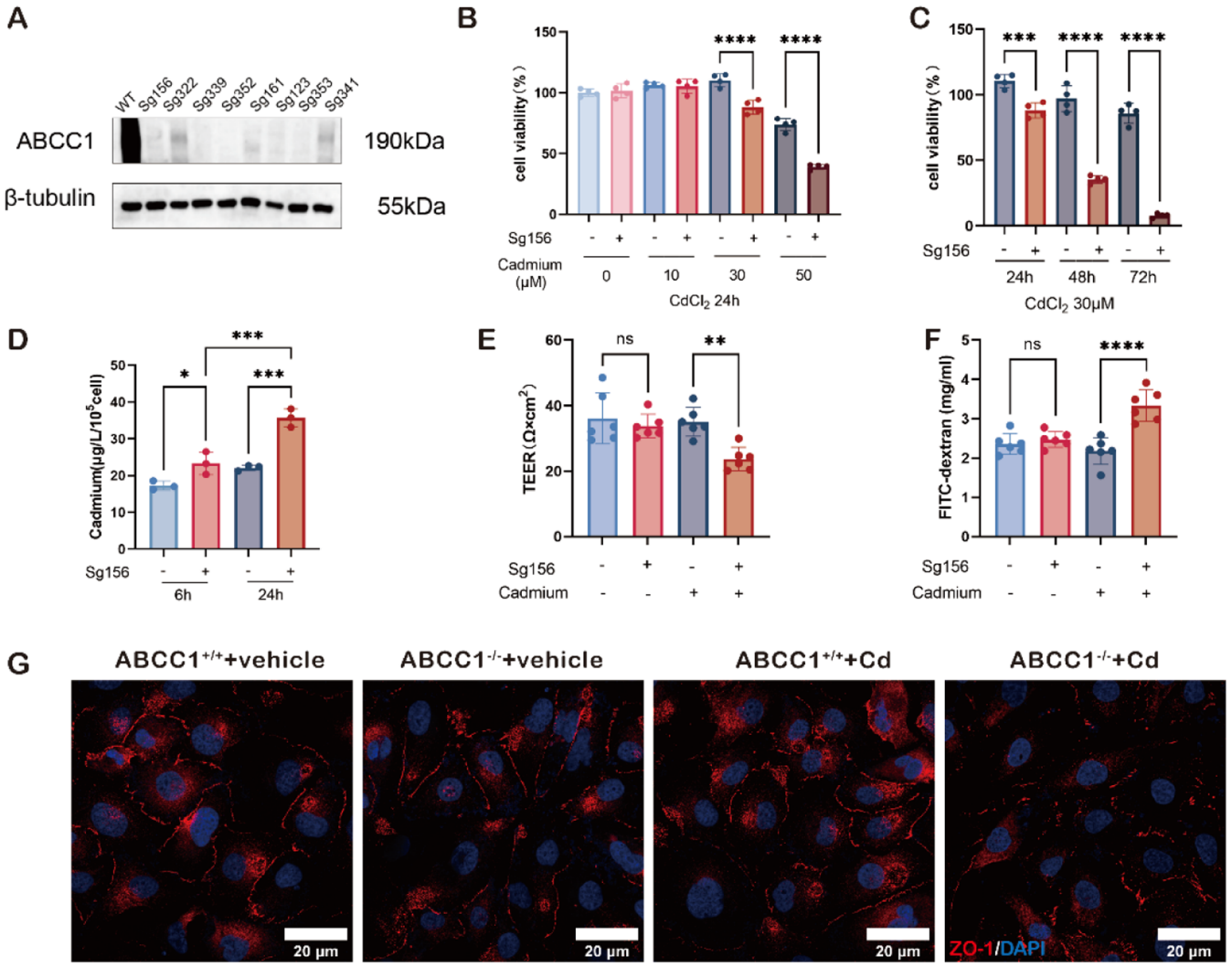


Figure 3. In vitro endothelial barrier model and cellular phenotypes. (A) Generation of ABCC1 knockout endothelial cells. $CdCl_2$ concentration-dependent (B) and time-dependent (C) cytotoxicities in endothelial cells. (D) Intracellular Cd accumulation in endothelial cells. (E) Endothelial cell TEER values. (F) The leakage of FITC-dextran into the lower chamber. (G) Representative immunofluorescence staining of ZO-1 in endothelial cells. Data are presented as mean ± SD; *p <0.05, **p <0.01, ***p <0.001, ****p <0.0001.

difference in blood Cd levels was observed between *Abcc1*$^{+/+}$ and *Abcc1*$^{-/-}$ mice (Figure 1D), indicating no direct impact on circulating Cd levels. *Abcc1*$^{-/-}$ mice consistently showed elevated Cd in brain, liver, and kidney tissues (Figure 1E−G). These data collectively underscore the critical role of ABCC1 in regulating in vivo Cd distribution and metabolism, suggesting its loss enhances susceptibility to Cd-induced hearing impairment and organ toxicity.

3.2. Cd Exposure Disrupts the BLB in *Abcc1*$^{-/-}$ Mice and Causes Cochlear SGNs and Presynaptic Ribbon Puncta Damage

ABCC1 is present in the microvascular endothelial cells of the cochlear stria vascularis,[24] a vital structural element of cochlear BLB. Our initial examination of BLB permeability, using intravenously administered FITC−dextran as a tracer, revealed leakage in the stria vascularis of Cd-exposed *Abcc1*$^{-/-}$ mice (Figure 2A and 2B). Transmission electron microscopic analysis further showed disrupted tight junctions between endothelial cells in Cd-treated *Abcc1*$^{-/-}$ mice, whereas junctions remained intact in *Abcc1*$^{+/+}$ mice (Figure 2C). These findings suggest that Cd compromises BLB integrity in *Abcc1*$^{-/-}$ mice by impairing the tight junctions of stria vascularis endothelial cells.

Our subsequent evaluation of key structural elements within the cochlear sensorineural system indicated those distinct outcomes. While immunofluorescence confocal imaging revealed that both *Abcc1*$^{+/+}$ and *Abcc1*$^{-/-}$ mice displayed normal inner and outer hair cell morphology with no hair cell loss (Figures 2D and 2E), further analysis of spiral ganglion neurons (SGNs) showed a marked reduction in the cell number in *Abcc1*$^{-/-}$ mice, with the loss predominantly occurring in the middle and basal turns of the cochlea (Figure 2F and 2G). Furthermore, when exposed to Cd, *Abcc1*$^{-/-}$ mice exhibited markedly fewer presynaptic ribbon puncta compared to their *Abcc1*$^{+/+}$ counterparts, with the most pronounced loss concentrated in the middle and basal turns (Figure 2H and 2I).

3.3. Cd Exposure Promotes Cd Accumulation and Impairs Barrier Function in *ABCC1*$^{-/-}$ Endothelial Cells

An in vitro endothelial cell model was established to unravel the molecular mechanisms of ABCC1, prompted by in vivo data showing higher Cd levels in *Abcc1*$^{-/-}$ mouse cochleae and the presence of ABCC1 in stria vascularis endothelial cells.

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581560/envhealth.6c00120.pdf by guest on 10 October 2026

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX

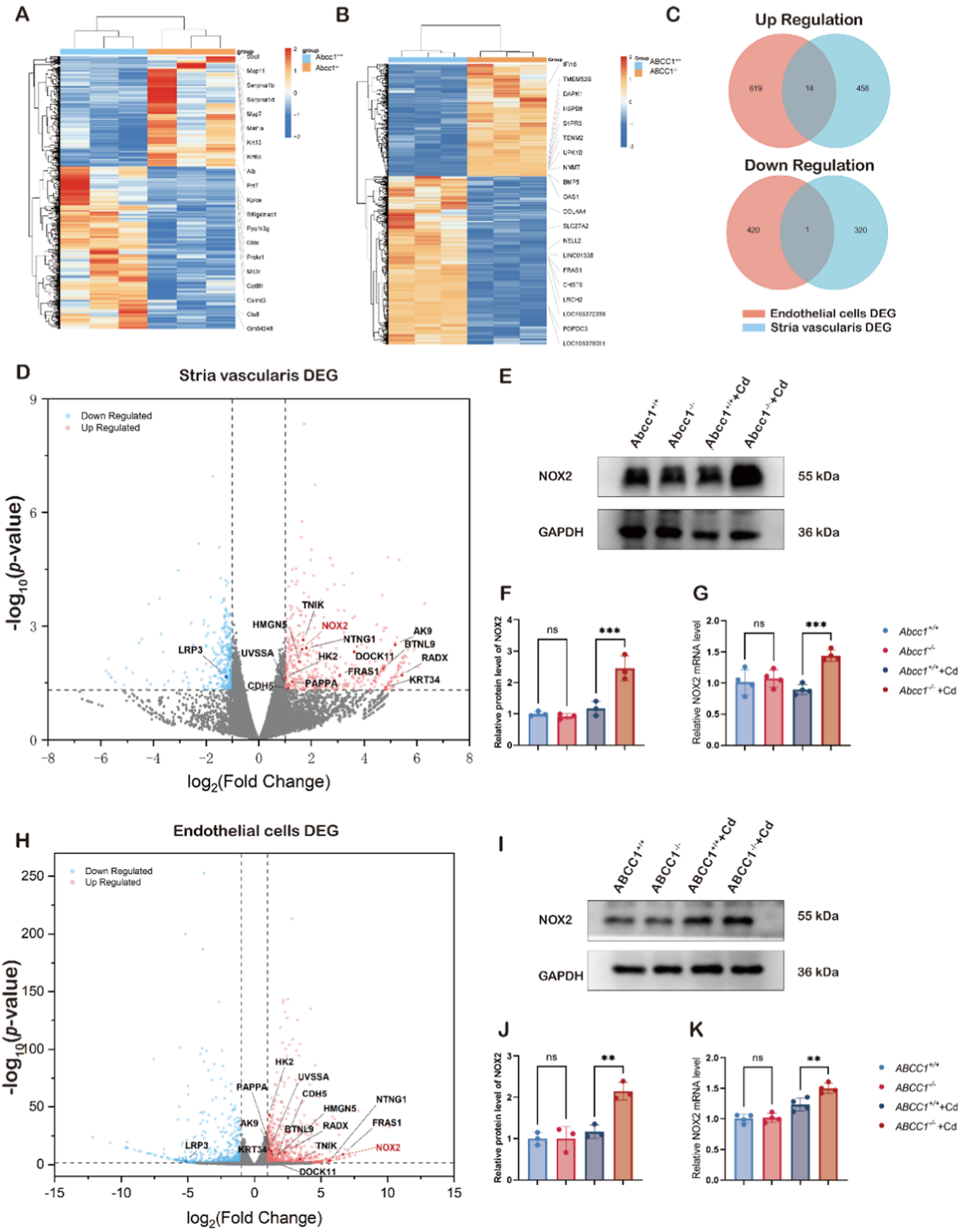


Figure 4. Effects of $CdCl_2$ exposure on the transcriptomes of cochlear and endothelial cells in ABCC1-deficient mice. Heatmap of DEGs in the stria vascularis (A) and endothelial cells (B). (C) Venn diagram of overlapping DEGs between stria vascularis and endothelial cells. (D) Volcano plot of DEGs in the cochlear stria vascularis. (E) Protein expression level of NOX2 in the stria vascularis. (F) Quantification of NOX2 protein band intensity in the stria vascularis. (G) RNA expression level of NOX2 in the stria vascularis. (H) Volcano plot of DEGs in endothelial cells. (I)

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581156/envhealth.6c00120.pdf by guest on 10 October 2026

Figure 4. continued

Protein expression level of NOX2 in endothelial cells. (J) Quantification of NOX2 protein band intensity in endothelial cells. (K) RNA expression level of NOX2 in endothelial cells. Data are presented as mean ± SD; $^{**}p$ <0.01, $^{***}p$ <0.001.

Using a lentivirus-mediated CRISPR/Cas9 system, stable HUVECs were generated via lentiviral transduction, followed by puromycin selection. Subsequently, ABCC1 knockout screening was conducted, leading to the successful isolation of single-cell-derived ABCC1 knockout HUVECs (Figure 3A).

Cellular viability was assessed using the CCK-8 assay, revealing that the impact of Cd on *ABCC1*$^{-/-}$ cells was contingent on both concentration and exposure duration. Following 24 h exposure to 30 μmol/L $CdCl_2$, *ABCC1*$^{-/-}$ cells exhibited significantly reduced viability compared to *ABCC1*$^{+/+}$ cells. Elevating the $CdCl_2$ concentration to 50 μmol/L further reduced *ABCC1*$^{-/-}$ cell viability to below 50% (Figure 3B), and extending the 30 μmol/L exposure to 48 h resulted in an even greater decline in *ABCC1*$^{-/-}$ cell viability (Figure 3C). These findings demonstrate the long-term protective role of ABCC1 in maintaining endothelial cell function and indicate that ABCC1 deficiency markedly compromises the capacity of cells to recuperate from Cd-induced injury, particularly during prolonged exposure.

Further analysis of intracellular Cd accumulation revealed that after 6 h of exposure to 30 μmol/L $CdCl_2$, *ABCC1*$^{-/-}$ cells accumulated significantly more Cd than their wild-type counterparts (1.3-fold higher after 6 h, increasing to 1.6-fold after 24 h) (Figure 3D). This heightened retention of Cd in endothelial cells, attributed to the absence of ABCC1, correlates with increased cellular toxicity. Given the vital role of endothelial cells in the structural and functional integrity of BLB, the endothelial barrier function was assessed. Cd exposure caused a marked reduction in TEER for *ABCC1*$^{-/-}$ cells compared to *ABCC1*$^{+/+}$ cells (p <0.001) (Figure 3E). This impairment was further substantiated by FITC−dextran permeability assays, which revealed a 1.5-fold increase in tracer flux across *ABCC1*$^{-/-}$ monolayers (p <0.001) (Figure 3F). Immunofluorescence staining for ZO-1, a tight junction protein, showed that Cd treatment led to discontinuous ZO-1 expression in *ABCC1*$^{-/-}$cells, signifying compromised tight junctions (Figure 3G).

3.4. ABCC1 Deficiency Mediates Cd-Induced NOX2 Upregulation in Cochlear Stria Vascularis and Endothelial Cells

To systematically characterize the transcriptional response of the cochlear stria vascularis and endothelial cells following $CdCl_2$ exposure under ABCC1-deficient conditions, stria vascularis tissues were collected from Cd-treated *Abcc1*$^{-/-}$ and *Abcc1*$^{+/+}$ mice, along with *ABCC1*$^{-/-}$ and *ABCC1*$^{+/+}$ endothelial cells. Total RNA was subsequently extracted from each sample for a high-throughput RNA sequencing analysis.

Heatmap visualization of transcriptomic data from the cochlear stria vascularis revealed that Cd exposure under ABCC1-deficient conditions substantially altered gene expression, indicating that ABCC1 plays a role in modulating the Cd-induced toxic response of the stria vascularis (Figure 4A). Similarly, *ABCC1*$^{-/-}$ cells exhibited a markedly distinct transcriptional profile compared to *ABCC1*$^{+/+}$ cells (Figure 4B). To identify common regulatory mechanisms between the in vivo and in vitro models, we performed an intersection analysis of the differentially expressed genes. This analysis identified 14 genes consistently upregulated after Cd exposure in both systems with only one gene commonly downregulated (Figure 4C). These coexpressed DEGs represent critical regulatory elements implicated in the heavy metal toxicity response under ABCC1-deficient conditions. Furthermore, their annotation in a volcano plot demonstrated strong statistical significance within the overall expression landscape (Figure 4D,H), highlighting their potential biological relevance and warranting further investigation.

Our findings reveal that the ABCC1 deficiency enhances the Cd-triggered elevation of NOX2, a crucial NADPH oxidase family member involved in oxidative stress. While baseline Western blot analysis showed no significant difference in NOX2 protein expression between the stria vascularis of *Abcc1*$^{-/-}$ and *Abcc1*$^{+/+}$ mice, $CdCl_2$ exposure led to a notable surge in the NOX2 protein and mRNA levels specifically in *Abcc1*$^{-/-}$ mice (Figure 4E−G). This suggests that ABCC1 insufficiency fosters Cd-provoked NOX2 upregulation, implying a potential contribution of NOX2 to the involvement in the oxidative damage response pathway in this context.

A similar trend was observed at the cellular level. Gene expression profiling in endothelial cells, particularly through volcano plot analysis, clearly highlighted NOX2 as one of the commonly upregulated genes following Cd exposure (Figure 4H). To verify the biological reliability of this finding, we assessed the NOX2 expression at both the protein and mRNA levels. Western blot analysis showed comparable NOX2 protein expression between *ABCC1*$^{-/-}$ and *ABCC1*$^{+/+}$ cells under basal conditions. However, upon $CdCl_2$ stimulation, NOX2 protein levels were markedly elevated in *ABCC1*$^{-/-}$ cells (Figure 4I and 4J). Similarly, Cd exposure also significantly increased NOX2 mRNA levels in *ABCC1*$^{-/-}$ cells (Figure 4K). These cellular results are consistent with the tissue-level observations, suggesting that the ABCC1 deficiency makes cells more vulnerable to Cd-induced oxidative stress by facilitating NOX2 production, thereby rendering cells more susceptible to Cd-induced damage.

3.5. Cd Induces Excessive ROS Accumulation and Redox Imbalance in the Cochleae and Endothelial Cells under ABCC1-Deficient Conditions

ABCC1 serves as a key transporter, facilitating the export of reduced glutathione (GSH) and oxidized glutathione (GSSG), which helps maintain intracellular redox balance.[27] Our transcriptomic analysis revealed that exposure to $CdCl_2$ significantly upregulated the oxidative stress-related gene NOX2 in the absence of ABCC1, suggesting that heightened oxidative stress could be a critical factor through which ABCC1 deficiency increases susceptibility to Cd-induced damage. To test this hypothesis, we further performed dihydroethidium (DHE) staining to quantify superoxide anion levels and determined the concentrations of GSH, GSSG, and their ratio, thereby systematically assessing the role of ABCC1 in antioxidant regulation.

Prior to $CdCl_2$ administration, ROS levels stayed minimal in the cochleae of both *Abcc1*$^{-/-}$ and *Abcc1*$^{+/+}$ mice, showing no notable disparity. However, following Cd exposure, *Abcc1*$^{-/-}$

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581156/envhealth.6c00120.pdf by guest on 10 October 2026

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX

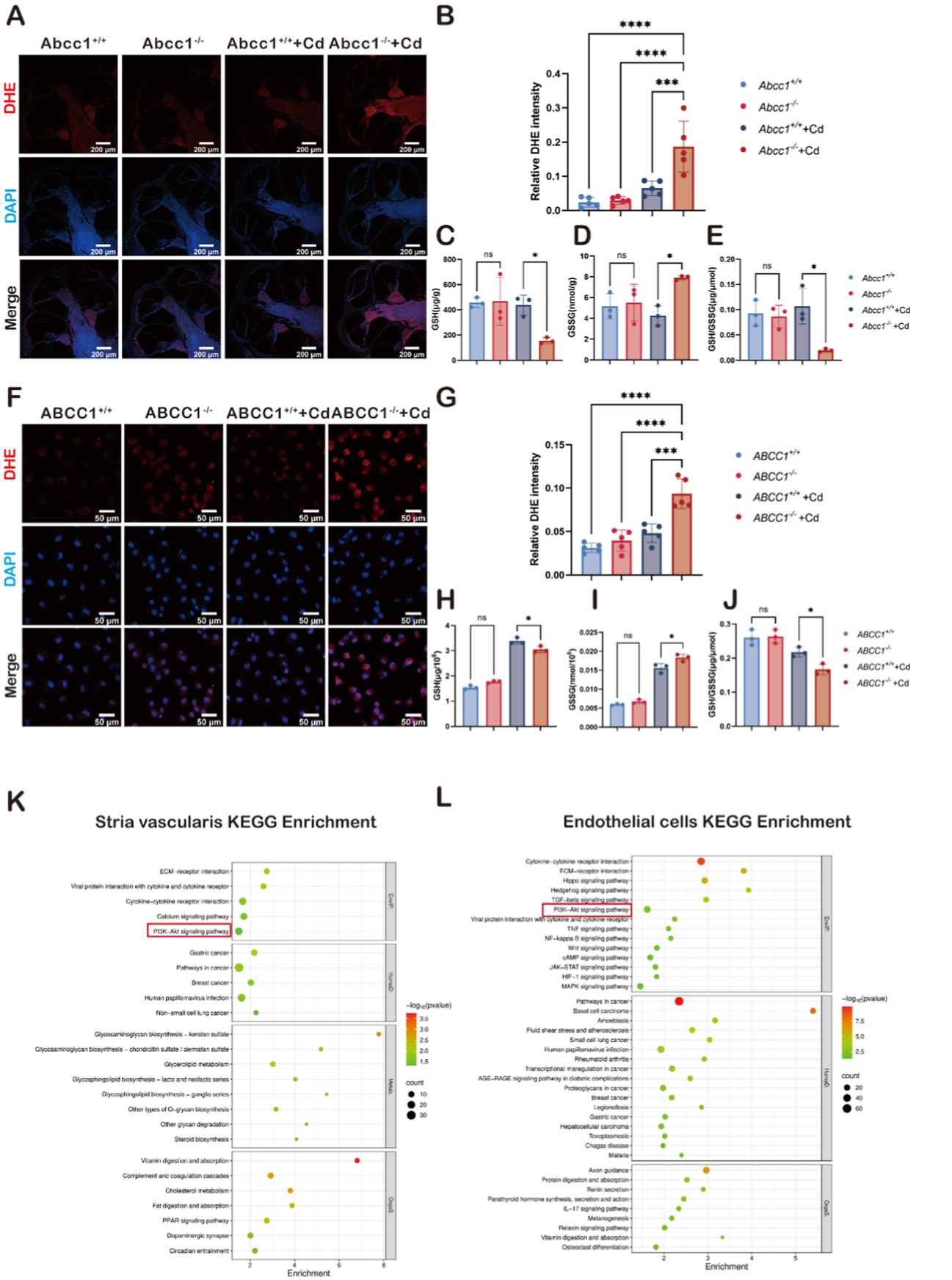


Figure 5. ROS and KEGG analysis in cells and cochlea. (A) Representative DHE immunofluorescence staining in the cochlea. (B) Quantification of DHE fluorescence intensity in the cochlea. (C) GSH content in the cochleae. (D) GSSG content in the cochleae. (E) GSH/GSSG ratio in the cochleae. (F) Representative DHE immunofluorescence staining in cells. (G) Quantification of DHE fluorescence intensity in cells. GSH content

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581560/envhealth.6c00120.pdf by guest on 10 October 2026

Figure 5. continued

(H), GSSG content (I), and GSH/GSSG ratio (J) in cells. (K) KEGG enrichment analysis of the stria vascularis. (L) KEGG enrichment analysis of endothelial cells. Data are presented as mean ± SD; *p <0.05, ***p <0.001, ****p <0.0001.

mice exhibited marked superoxide anion accumulation, as evidenced by a pronounced increase in the DHE fluorescence signal, predominantly localized within the stria vascularis and spiral ganglion regions. This finding indicates that in the absence of ABCC1, the cochlear tissue is more susceptible to Cd-induced oxidative stress (Figure 5A). Quantitative analysis further confirmed this trend, showing significantly higher fluorescence intensity in the cochleae of *Abcc1*$^{-/-}$ mice compared with *Abcc1*$^{+/+}$ controls (Figure 5B).

Investigation of the glutathione system revealed changes consistent with ROS accumulation. In the cochleae of *Abcc1*$^{-/-}$ mice, Cd exposure led to a marked decrease in GSH (Figure 5C) and a concurrent increase in GSSG, reflecting extensive conversion of GSH to GSSG under oxidative pressure (Figure 5D), pointing to diminished antioxidant stores and increased oxidative stress. More importantly, the GSH/GSSG ratio was significantly decreased, demonstrating that the cochlear tissue had shifted toward a redox-imbalanced state dominated by oxidative stress (Figure 5E).

Similarly, compared with *ABCC1*$^{+/+}$ cells, *ABCC1*$^{-/-}$ cells, though maintaining stable ROS levels under control conditions, displayed pronounced ROS accumulation when exposed to Cd, as evidenced by markedly increased fluorescence intensity (Figure 5Fand 5G). These cellular changes mirrored those observed in tissues, with reduced GSH (Figure 5H), elevated GSSG (Figure 5I), and a decreased GSH/GSSG ratio (Figure 5J) in *ABCC1*$^{-/-}$ cells, indicating a significant impairment of their antioxidant defense capacity.

3.6. Cd Suppresses PI3K-AKT Signaling in the Cochleae and Endothelial Cells under ABCC1-Deficient Conditions

To further explore the molecular mechanisms underlying the observed phenotypic changes, KEGG pathway enrichment analysis was performed on the RNA-seq data. The results revealed that the PI3K-AKT signaling pathway was among the significantly enriched pathways in both cochlear stria vascularis tissue and endothelial cells (Figure 5K and 5L). As a key regulator of cellular stress responses, survival, and antioxidant defense, this pathway may be involved in mediating oxidative damage responses under the ABCC1-deficient conditions. These findings suggest that ABCC1 may contribute to antioxidant protection by modulating the PI3K-AKT pathway, thereby exerting a protective role during Cd-induced oxidative stress.

To verify the role of PI3K-AKT signaling during oxidative stress, we assessed the protein expression of key PI3K-AKT pathway components in the cochlear tissue and endothelial cells. Our findings indicate that ABCC1 deficiency significantly impairs AKT activation under oxidative stress, evidenced by a marked reduction in p-AKT expression (Figure 6A and 6D), and the p-AKT/AKT ratio in *Abcc1*$^{-/-}$ mouse cochleae (Figure 6E) after Cd exposure, while PI3K and total AKT levels remained unchanged (Figure 6B and 6C). Conversely, under basal conditions without Cd exposure, ABCC1 deficiency did not affect PI3K-AKT signaling, as PI3K, total AKT, and p-AKT levels were comparable between *Abcc1*$^{-/-}$ and *Abcc1*$^{+/+}$ mice.

Similar findings were observed in the endothelial cells. In the absence of Cd treatment, no significant differences were observed in the expression of PI3K, total AKT, or p-AKT between *ABCC1*$^{+/+}$ and *ABCC1*$^{-/-}$ cells. Following Cd exposure, however, p-AKT levels were significantly decreased in *ABCC1*$^{-/-}$ cells (Figure 6F,I), while the expression of PI3K and total AKT remained unchanged (Figure 6G,H). Quantification confirmed a significantly lower p-AKT/AKT ratio in the *ABCC1*$^{-/-}$ group compared to the *ABCC1*$^{+/+}$ group (Figure 6J). Collectively, these findings reinforce that ABCC1 deficiency suppresses AKT activation under Cd-induced oxidative stress conditions.

3.7. NOX2 Inhibition Attenuates Cd-Induced Suppression of the ROS/PI3K-AKT Axis and Endothelial Barrier Dysfunction in *ABCC1*$^{-/-}$ Cells

To further elucidate the critical role of NOX2 in oxidative stress activation and functional impairment in *ABCC1*$^{-/-}$ cells, we conducted experiments using the specific NOX2 inhibitor GSK2795039 (10 μmol/L). The NADP$^+$/NADPH ratio, a sensitive indicator of redox metabolic status, was measured to assess cellular reducing capacity. Upon Cd exposure, the NADP$^+$/NADPH ratio was significantly elevated in *ABCC1*$^{-/-}$ cells, indicating reduced cellular reducing power (Figure 6K). Notably, cotreatment with GSK2795039 restored the ratio in *ABCC1*$^{-/-}$ cells to levels comparable with *ABCC1*$^{+/+}$ cells, highlighting the central role of NOX2 in mediating oxidative stress. To determine whether NOX2 contributes to Cd-induced reductions in cell viability, we conducted CCK-8 assays to assess the effect of GSK2795039 intervention. While *ABCC1*$^{-/-}$ cells exhibited a substantial decrease in viability after Cd exposure, this decline was largely mitigated by NOX2 inhibition (Figure 6L). This outcome confirms NOX2 as a significant contributor to Cd-induced cytotoxicity in the absence of ABCC1.

To determine whether NOX2 mediates ROS accumulation, we employed DHE staining to evaluate the effects of GSK2795039 treatment. Baseline ROS levels were similar between *ABCC1*$^{-/-}$ and *ABCC1*$^{+/+}$ cells without Cd exposure, indicating that the NOX2 inhibitor alone does not disturb cellular redox homeostasis. However, Cd exposure significantly boosted superoxide levels in *ABCC1*$^{-/-}$ cells. Co-treatment with GSK2795039 effectively reduced ROS fluorescence in *ABCC1*$^{-/-}$ cells, indicating that NOX2 plays a pivotal role in Cd-induced ROS accumulation (Figure 7A and 7B). In terms of antioxidant metabolism, Cd exposure significantly decreased GSH levels and increased GSSG levels in *ABCC1*$^{-/-}$ cells, signifying rapid antioxidant loss and an oxidized cellular state. In contrast, cotreatment with GSK2795039 markedly attenuated GSH depletion and GSSG accumulation, thereby preserving the GSH/GSSG ratio. These findings suggest that NOX2 inhibition alleviates Cd-induced redox imbalance (Figure 7C−E).

Building on prior KEGG analysis linking the PI3K-AKT pathway to Cd-induced stress responses, we further investigated whether this pathway is regulated by NOX2. Our assessment of PI3K-AKT protein expression under various conditions revealed that GSK2795039 treatment alone did not

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581560/envhealth.6c00120.pdf by guest on 10 October 2026

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX

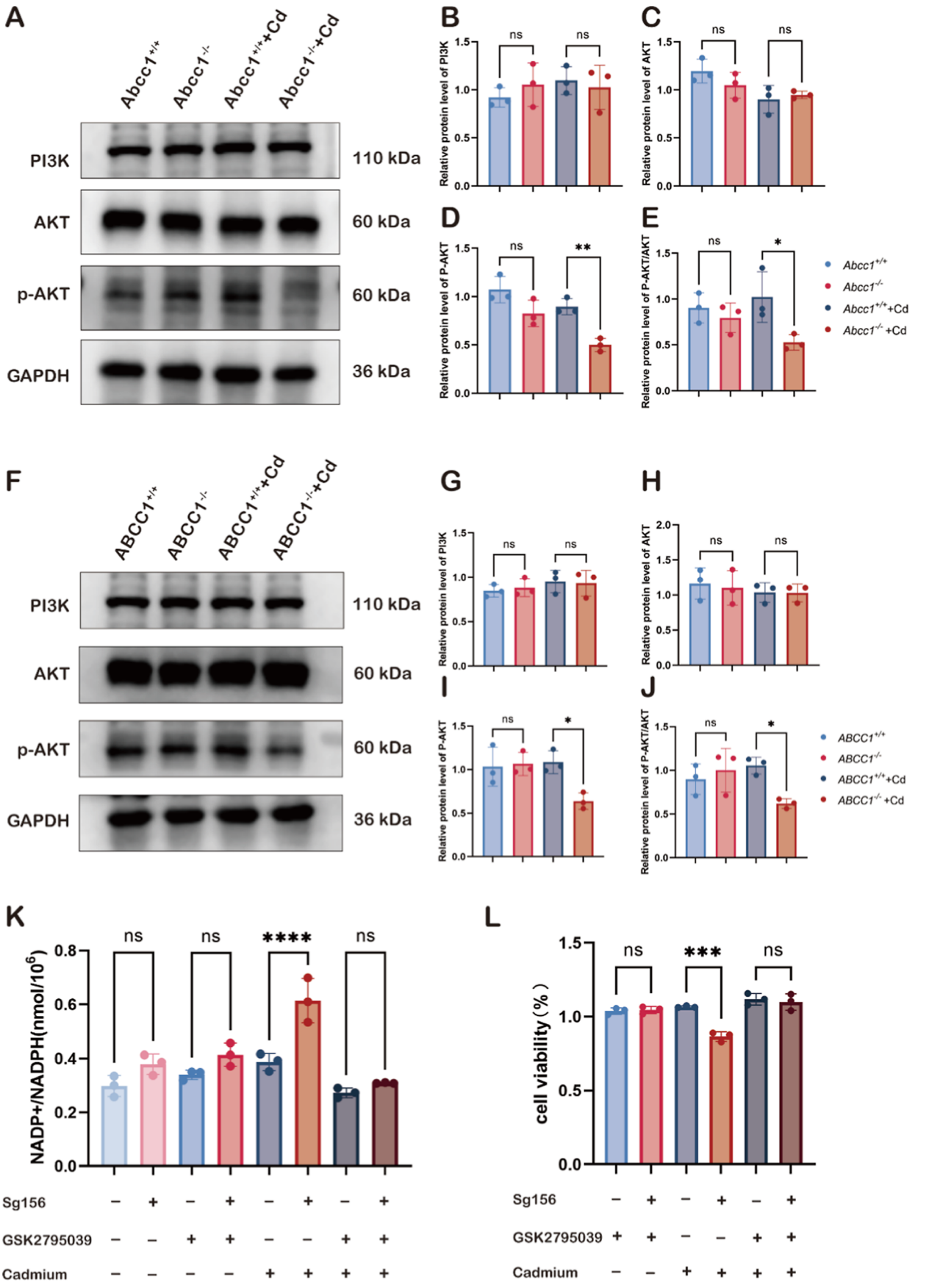


Figure 6. Expression levels of key proteins in the PI3K-AKT pathway. (A) Protein expression levels of PI3K, AKT, and P-AKT in the cochleae. Quantification of PI3K (B), AKT (C), and P-AKT (D) protein band intensities in the cochleae. (E) Quantification of the P-AKT/AKT ratio in the cochleae. (F) Protein expression levels of PI3K, AKT, and P-AKT in endothelial cells. (G–I) Quantification of PI3K, AKT, and P-AKT protein

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581156/envhealth.6c00120.pdf by guest on 10 October 2026

Figure 6. continued

band intensities in cells. (J) Quantification of the P-AKT/AKT ratio in cells. (K) $NADP^+$/NADPH ratio in cells. (L) Cell viability following GSK2795039 treatment. Data are presented as mean ± SD; *p <0.05, **p <0.01, ***p <0.001, ****p <0.0001.

alter basal p-AKT levels in $ABCC1^{-/-}$ cells, nor did it affect total PI3K or AKT, confirming that the NOX2 inhibitor does not influence baseline pathway activity. Cd exposure led to a marked decrease in p-AKT expression within $ABCC1^{-/-}$ cells. However, cotreatment with the NOX2 inhibitor and Cd restored p-AKT expression to baseline levels and preserved the p-AKT/AKT ratio (Figure 7F–J). This evidence points to NOX2–ROS-dependent oxidative stress as a mediator for PI3K-AKT signaling suppression in $ABCC1^{-/-}$ cells.

To further assess whether NOX2 inhibition improves endothelial barrier integrity in $ABCC1^{-/-}$ cells, we examined the architecture and functionality of tight junctions. Treatment with GSK2795039 alone resulted in continuous and structurally intact ZO-1 staining. In contrast, Cd exposure notably fragmented ZO-1 and caused discontinuous expression in $ABCC1^{-/-}$ cells, signifying a compromised barrier structure. Co-treatment with the NOX2 inhibitor significantly alleviated these structural abnormalities, restoring a more continuous ZO-1 appearance (Figure 7K). Consistent with this, TEER measurements showed no significant difference between $ABCC1^{-/-}$ and control cells under GSK2795039 treatment alone. However, Cd exposure led to a marked reduction in TEER in $ABCC1^{-/-}$ cells, which was substantially rescued by NOX2 inhibition (Figure 7L). To further validate these findings, a FITC-dextran permeability assay was conducted. Cd exposure led to significantly increased FITC–dextran flux to the lower chamber of $ABCC1^{-/-}$ cell cultures. Co-treatment with GSK2795039 markedly reduced FITC-dextran leakage, in line with TEER results, demonstrating that NOX2 inhibition effectively mitigates barrier dysfunction in $ABCC1^{-/-}$ cells (Figure 7M). Taken together, these results highlight the critical role of NOX2 in Cd-induced oxidative stress, PI3K-AKT pathway suppression, and barrier disruption. Importantly, NOX2 inhibition effectively alleviates Cd-induced damage in the context of ABCC1 deficiency, further supporting the involvement of the NOX2-ROS/PI3K-AKT axis in mediating endothelial injury.

4. DISCUSSION

Our study, using both $Abcc1^{-/-}$ mice and $ABCC1^{-/-}$ HUVECs models, demonstrates that ABCC1 plays a critical protective role against Cd-induced cochlear injury by maintaining BLB integrity and redox homeostasis. ABCC1 deficiency resulted in increased Cd accumulation in the cochlea and various tissues, induced structural stria vascularis damage, upregulated NOX2 expression, promoted ROS accumulation, and suppressed PI3K-AKT signaling. These changes ultimately result in SGN and presynaptic ribbon puncta injury, accompanied by hearing loss. Pharmacological inhibition of NOX2 partially reversed these effects, supporting the functional role of NOX2-mediated oxidative stress in this process.

ABCC1, mediating the transmembrane export of a broad range of endogenous and exogenous substrates, including drugs, cysteinyl leukotrienes, and glutathione conjugates, is vital for organ protection against heavy metal toxicity.[27,28] In this study, ABCC1 deficiency led to increased Cd accumulation in multiple organs without altering circulating Cd levels, suggesting that ABCC1 primarily regulates tissue-specific Cd distribution. Its role is particularly critical in the cochlea, where ABCC1 is heavily expressed in the endothelial cells of the stria vascularis—an essential structural component of the BLB.[24,29,30] The localization of ABCC1 within BLB endothelial cells endows it with a "gatekeeping" role, as evidenced by the significantly increased BLB permeability and structural disruption observed in $Abcc1^{-/-}$ mice following Cd exposure. In vitro experiments further demonstrated that Cd exposure in $ABCC1^{-/-}$ endothelial cells led to disruption of tight junctions, which is critical for barrier integrity. Their disruption facilitates the rapid entry of Cd into the endolymphatic space, thereby triggering a cascade of downstream cellular damage.[31,32] These findings emphasize the indispensable role of BLB integrity, with ABCC1 as a pivotal protector, against cochlear heavy metal toxicity. However, as global Abcc1 knockout mice were used, systemic effects cannot be excluded, and endothelial-specific contributions require further validation using conditional knockout models.

In light of these findings, it is important to consider the environmental relevance of the exposure paradigm used in this study. Although the Cd concentration employed exceeds typical human exposure levels, Cd toxicity is characterized by its long biological half-life and progressive accumulation in tissues. Therefore, the current model is intended to reproduce cumulative tissue burden within a limited experimental time frame. Notably, Cd accumulation was observed in the cochlear tissue despite similar blood Cd levels, supporting the concept that tissue burden rather than circulating Cd determines the toxicity. From this perspective, the current model represents an accelerated-accumulation scenario.

Alongside BLB disruption, we also observed significant alterations in auditory neural structures. $Abcc1^{-/-}$ mice exhibited pronounced SGN loss and reduced presynaptic ribbon puncta following Cd exposure. These observations point to neural injury as the primary cause of hearing impairment due to Cd toxicity. Previous studies have demonstrated that Cd induces neurotoxicity through multiple mechanisms, including disruption of calcium homeostasis, excessive reactive ROS generation, and mitochondrial damage.[33–35] Our results further indicate that ABCC1 deficiency could increase susceptibility to these toxic effects, rendering SGNs and presynaptic ribbon puncta more vulnerable to Cd-induced injury. In addition, increased BLB permeability may facilitate Cd entry into the cochlear neural microenvironment, thereby exacerbating neuronal damage. However, intrinsic neuronal mechanisms were not directly assessed, and the observed neural damage is likely secondary to BLB disruption and Cd accumulation.

Notably, despite evident injury to the stria vascularis, SGNs, and presynaptic ribbon puncta, no obvious hair cell loss was detected in this study. This selective vulnerability may be related to the localization of ABCC1 in stria vascularis endothelial cells and its role in maintaining BLB integrity, which may preferentially affect vascular and neural compartments under Cd exposure. In contrast, previous studies of Cd ototoxicity have mainly used cochlear explants, in which cochlear tissues are directly exposed to Cd, and the protective

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581156/envhealth.6c00120.pdf by guest on 10 October 2026

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX–XXX

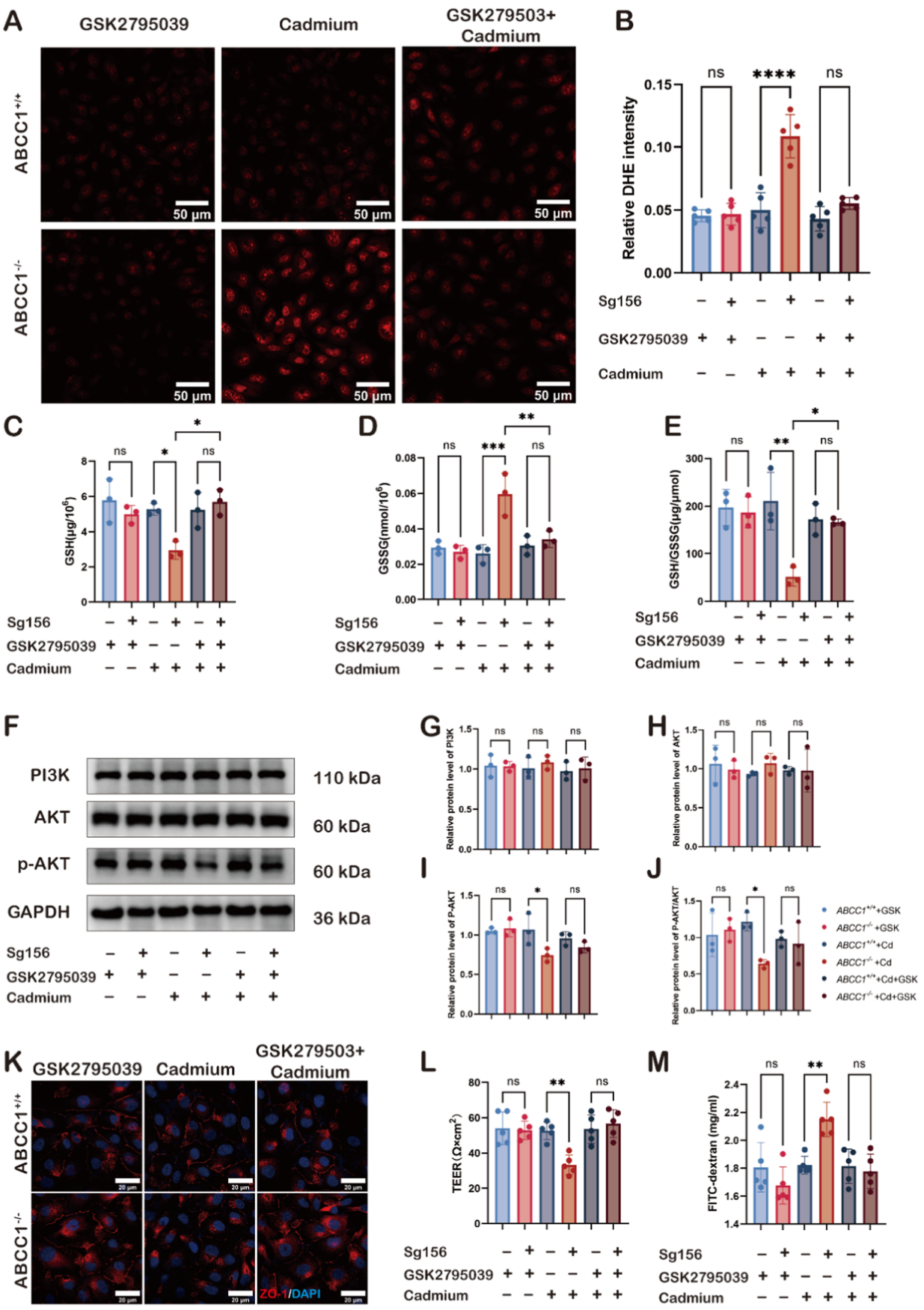


Figure 7. Cellular phenotypes and PI3K-AKT pathway changes following GSK2795039 treatment. (A) Representative DHE staining in endothelial cells. (B) Quantification of DHE fluorescence intensity in cells. GSH levels (C), GSSG levels (D), and GSH/GSSG ratio (E) in cells. (F) Protein expression levels of PI3K, AKT, and P-AKT in endothelial cells. Quantification of PI3K (G), AKT (H), and *p*-AKT (I) protein band intensities in

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581560/envhealth.6c00120.pdf by guest on 10 October 2026

Figure 7. continued

cells. (J) Quantification of the P-AKT/AKT ratio in cells. (K) Representative ZO-1 immunofluorescence staining in endothelial cells. (L) TEER values of endothelial cells. (M) FITC-dextran leakage into the lower chamber. Data are presented as mean ± SD; *p <0.05, **p <0.01, ***p <0.001, ****p <0.0001.

role of the BLB is not fully represented. This difference in experimental context may partly explain why vascular and neural injuries predominated in our in vivo model, whereas overt hair cell loss was not observed. Nevertheless, subtle functional or ultrastructural alterations in hair cells cannot be excluded and require further investigation.

Building on this, we observed that ABCC1 deficiency amplified Cd-induced oxidative stress, accompanied by increased NOX2 expression and ROS accumulation.[36−38] In ABCC1-deficient mice and cells, Cd exposure led to depletion of GSH, accumulation of GSSG, and a reduced GSH/GSSG ratio, indicating impaired redox homeostasis. Given the role of ABCC1 in glutathione transport, its deficiency may disrupt intracellular redox balance and increase vulnerability to oxidative stress under Cd exposure. Furthermore, treatment with the GSK2795039 reversed these damaging effects, further confirming the central role played by NOX2 in this pathological process. These findings align with previous studies reporting the involvement of NOX2 in Cd-induced toxicity in hepatic tissue and neural cells, suggesting that NOX2 may represent a shared therapeutic target across different organs affected by Cd exposure.[39,40] However, the present study did not establish a direct regulatory relationship between ABCC1 and NOX2. NOX2 expression was comparable between ABCC1-deficient and control models under basal conditions, whereas significant NOX2 upregulation occurred following Cd exposure. Therefore, it remains unclear whether NOX2 activation is driven primarily by increased Cd accumulation resulting from ABCC1 deficiency or whether ABCC1 also exerts additional regulatory effects on NOX2 expression or activity. Further studies are needed to clarify this mechanism.

We further notice that oxidative stress intersected with intracellular signaling cascades, most notably the PI3K-AKT pathway. As a fundamental signaling axis, the PI3K-AKT pathway is essential for cellular adaptation to external stressors and is deeply involved in processes such as metabolism, cell proliferation, survival, and apoptosis while also being a key modulator of several antioxidant defense systems.[41−43] Our study demonstrated that Cd exposure significantly impaired AKT phosphorylation in the cochlear tissue and endothelial cells lacking ABCC1, suggesting a decrease in pathway activity. However, cotreatment with the GSK2795039 restored p-AKT expression and the p-AKT/AKT ratio, indicating that oxidative stress serves as an upstream driver of this pathway suppression. Interestingly, the PI3K-AKT pathway can also regulate downstream antioxidant factors, such as Nrf2 and HO-1, which participate in ROS clearance.[44−46] Therefore, ABCC1 deficiency leads to upregulation of NOX2 expression after Cd treatment, resulting in a significant accumulation of ROS within the cell. Excess ROS not only exacerbates oxidative damage but also inhibits AKT phosphorylation, potentially weakening the antioxidant defense functions mediated by the PI3K-AKT pathway, further intensifying oxidative stress and creating a self-amplifying feedback loop of oxidative damage.

These results propose that boosting ABCC1 activity or targeted NOX2 inhibition could offer therapeutic approaches to Cd-induced hearing loss. Moreover, genetic epidemiological studies are required in the future to determine whether individuals with impaired ABCC1 function are more vulnerable to heavy-metal-induced ototoxicity.

5. CONCLUSION

This pioneering study is the first to establish the vital protective role of ABCC1 against Cd-induced cochlear injury, proposing that it maintains BLB homeostasis by modulating the NOX2−ROS/PI3K-AKT signaling pathway, thereby curtailing the cascade amplification of Cd toxicity. These findings not only reveal the molecular basis of cochlear responses to heavy metal exposure but also identify promising therapeutic targets for ototoxicity prevention and treatment of ototoxicity. Clinically, ABCC1 expression levels or functional status could serve as a biomarker for predicting individual susceptibility to environmental Cd exposure, offering a theoretical foundation for the development of personalized intervention strategies. Future research should further investigate the broader relevance of the ABCC1 regulatory network in other ototoxicity models and evaluate its translational potential and drug-target feasibility, paving the way for novel approaches to mitigate ototoxic substance-associated hearing impairment.

■ ASSOCIATED CONTENT

Data Availability Statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/envhealth.6c00120.

Baseline and Cd-exposure ABR assessments, reagents and assay kits, antibodies used for immunofluorescence and Western blot analyses, and primer sequences used for qRT-PCR assays (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Tianhan Kai − *Xiangya School of Public Health, Central South University, Changsha, Hunan 410013, China; Furong Laboratory, Changsha, Hunan 410008, China;* orcid.org/0000-0001-5667-1426; Email: th_kai@csu.edu.cn

Lingyun Mei − *Department of Otolaryngology-Head and Neck Surgery, Xiangya Hospital, Central South University, Changsha, Hunan 410008, China; Otolaryngology Major Disease Research Key Laboratory of Hunan Province, Changsha, Hunan 410008, China; National Clinical Research Center for Geriatric Disorders (Xiangya Hospital), Changsha, Hunan 410008, China;* orcid.org/0000-0003-3034-3322; Email: entmly@163.com

Authors

Shuai Zhang − *Department of Otolaryngology-Head and Neck Surgery, Xiangya Hospital, Central South University,*

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581156/envhealth.6c00120.pdf by guest on 10 October 2026

Changsha, Hunan 410008, China; Otolaryngology Major Disease Research Key Laboratory of Hunan Province, Changsha, Hunan 410008, China; National Clinical Research Center for Geriatric Disorders (Xiangya Hospital), Changsha, Hunan 410008, China

Yongjia Chen − *Department of Otolaryngology-Head and Neck Surgery, Xiangya Hospital, Central South University, Changsha, Hunan 410008, China; Otolaryngology Major Disease Research Key Laboratory of Hunan Province, Changsha, Hunan 410008, China; National Clinical Research Center for Geriatric Disorders (Xiangya Hospital), Changsha, Hunan 410008, China*

Yijiang Bai − *Department of Otolaryngology-Head and Neck Surgery, Xiangya Hospital, Central South University, Changsha, Hunan 410008, China; Otolaryngology Major Disease Research Key Laboratory of Hunan Province, Changsha, Hunan 410008, China; National Clinical Research Center for Geriatric Disorders (Xiangya Hospital), Changsha, Hunan 410008, China*

Bo Pang − *Department of Otolaryngology-Head and Neck Surgery, Xiangya Hospital, Central South University, Changsha, Hunan 410008, China; Otolaryngology Major Disease Research Key Laboratory of Hunan Province, Changsha, Hunan 410008, China; National Clinical Research Center for Geriatric Disorders (Xiangya Hospital), Changsha, Hunan 410008, China*

Mengzhu Jiang − *Department of Otolaryngology-Head and Neck Surgery, Xiangya Hospital, Central South University, Changsha, Hunan 410008, China; Otolaryngology Major Disease Research Key Laboratory of Hunan Province, Changsha, Hunan 410008, China; National Clinical Research Center for Geriatric Disorders (Xiangya Hospital), Changsha, Hunan 410008, China*

Zequn Nie − *Department of Otolaryngology-Head and Neck Surgery, Xiangya Hospital, Central South University, Changsha, Hunan 410008, China; Otolaryngology Major Disease Research Key Laboratory of Hunan Province, Changsha, Hunan 410008, China; National Clinical Research Center for Geriatric Disorders (Xiangya Hospital), Changsha, Hunan 410008, China*

Huping Huang − *Department of Otolaryngology-Head and Neck Surgery, Xiangya Hospital, Central South University, Changsha, Hunan 410008, China; Otolaryngology Major Disease Research Key Laboratory of Hunan Province, Changsha, Hunan 410008, China; National Clinical Research Center for Geriatric Disorders (Xiangya Hospital), Changsha, Hunan 410008, China*

Lu Wang − *Department of Otolaryngology-Head and Neck Surgery, Xiangya Hospital, Central South University, Changsha, Hunan 410008, China; Otolaryngology Major Disease Research Key Laboratory of Hunan Province, Changsha, Hunan 410008, China; National Clinical Research Center for Geriatric Disorders (Xiangya Hospital), Changsha, Hunan 410008, China*

Jie Ling − *Medical Functional Experiment Center, Xiangya School of Basic Medical Sciences, Central South University, Changsha, Hunan 410078, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/envhealth.6c00120

Author Contributions

[¶]L.M. and T.K. have contributed equally to this work.

Notes

Ethics approval and consent to participate: The care and experiments of animals were approved by the Ethics Committees of Central South University (Approval #: CSU-2022−0143). The related procedures were in accordance with the Association for Assessment and Accreditation of Laboratory Animal Care guidelines and conformed to the Animal Protection Act of China.

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (Grant No. 82171154), 2025 Hunan Province Key R&D Program Project (Grant No. 2025JK2141), and the Central South University Research Programme of Advanced Interdisciplinary Studies (Grant No. 2023QYJC013).

■ REFERENCES

(1) Chadha, S.; Kamenov, K.; Cieza, A. The world report on hearing, 2021. *Bull. World Health Organ.* **2021**, *99* (4), 242.

(2) Lieu, J. E. C.; Kenna, M.; Anne, S.; Davidson, L. Hearing Loss in Children: A Review. *JAMA* **2020**, *324* (21), 2195−2205.

(3) Cunningham, L. L.; Tucci, D. L. Hearing Loss in Adults. *N. Engl. J. Med.* **2017**, *377* (25), 2465−2473.

(4) Zhang, J.; Gong, T.; Chen, P.; Zhu, J.; Huang, S.; Li, Y.; Li, G.; Zhang, Q.; Duan, M.; Song, Q.; et al. Connexin30-deficient mice increase susceptibility to noise via redox and lactate imbalances. *Free Radic. Biol. Med.* **2024**, *225*, 641−653.

(5) Bae, S. H.; Yoo, J. E.; Hong, J. W.; Park, H. R.; Noh, B.; Kim, H.; Kang, M.; Hyun, Y.-M.; Gee, H. Y.; Choi, J. Y.; et al. LCCL peptide cleavage after noise exposure exacerbates hearing loss and is associated with the monocyte infiltration in the cochlea. *Hear. Res.* **2021**, *412*, 108378.

(6) Rajagopalan, S.; Landrigan, P. J. Pollution and the Heart. *N. Engl. J. Med.* **2021**, *385* (20), 1881−1892.

(7) Järup, L. Hazards of heavy metal contamination. *Br. Med. Bull.* **2003**, *68*, 167−182.

(8) Satarug, S.; Baker, J. R.; Urbenjapol, S.; Haswell-Elkins, M.; Reilly, P. E. B.; Williams, D. J.; Moore, M. R. A global perspective on cadmium pollution and toxicity in non-occupationally exposed population. *Toxicol. Lett.* **2003**, *137* (1−2), 65−83.

(9) Pamphlett, R.; Bishop, D. P.; Kum Jew, S.; Doble, P. A. Age-related accumulation of toxic metals in the human locus ceruleus. *PLoS One* **2018**, *13* (9), No. e0203627.

(10) Choi, Y.-H.; Hu, H.; Mukherjee, B.; Miller, J.; Park, S. K. Environmental cadmium and lead exposures and hearing loss in U.S. adults: the National Health and Nutrition Examination Survey, 1999 to 2004. *Environ. Health Perspect.* **2012**, *120* (11), 1544−1550.

(11) Ding, D.; Zhang, J.; Jiang, H.; Xuan, W.; Qi, W.; Salvi, R. Some Ototoxic Drugs Destroy Cochlear Support Cells Before Damaging Sensory Hair Cells. *Neurotox. Res.* **2020**, *37* (3), 743−752.

(12) Li, Q.; Wang, L.; Ji, D.; Yu, W.; Zhang, Y.; Xiang, Y.; Zhou, C.; Wang, L.; Deng, P.; Pi, H.; et al. Metformin attenuates cadmium-induced degeneration of spiral ganglion neuron via restoring autophagic flux in primary culture. *J. Inorg. Biochem.* **2022**, *234*, 111901.

(13) Kim, S.-J.; Myung, N.-Y.; Shin, B.-G.; Lee, J.-H.; So, H.-S.; Park, R.-K.; Um, J.-Y.; Hong, S.-H. Protective effect of a Chrysanthemum indicum containing formulation in cadmium-induced ototoxicity. *Am. J. Chin. Med.* **2011**, *39* (3), 587−600.

(14) Li, M.; Mei, L.; He, C.; Chen, H.; Cai, X.; Liu, Y.; Tian, R.; Tian, Q.; Song, J.; Jiang, L.; et al. Extrusion pump ABCC1 was first linked with nonsyndromic hearing loss in humans by stepwise genetic analysis. *Genet. Med.* **2019**, *21* (12), 2744−2754.

(15) Hanafy, A. S.; Lamprecht, A.; Dietrich, D. Local perfusion of capillaries reveals disrupted beta-amyloid homeostasis at the blood-

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581560/envhealth.6c00120.pdf by guest on 10 October 2026

brain barrier in Tg2576 murine Alzheimer's model. *Fluids Barriers CNS* **2023**, *20* (1), 85.
(16) Qian, Y. M.; Song, W. C.; Cui, H.; Cole, S. P.; Deeley, R. G. Glutathione stimulates sulfated estrogen transport by multidrug resistance protein 1. *J. Biol. Chem.* **2001**, *276* (9), 6404−6411.
(17) Hooijberg, J. H.; Jansen, G.; Assaraf, Y. G.; Kathmann, I.; Pieters, R.; Laan, A. C.; Veerman, A. J. P.; Kaspers, G. J. L.; Peters, G. J. Folate concentration dependent transport activity of the Multidrug Resistance Protein 1 (ABCC1). *Biochem. Pharmacol.* **2004**, *67* (8), 1541−1548.
(18) Xu, X.; Wang, T.; Sun, M.; Bai, Y.; Fu, C.; Zhang, L.; Hu, X.; Hagist, S. Management principles for heavy metal contaminated farmland based on ecological risk-A case study in the pilot area of Hunan province, China. *Sci. Total Environ.* **2019**, *684*, 537−547.
(19) Zou, M.; Zhou, S.; Zhou, Y.; Jia, Z.; Guo, T.; Wang, J. Cadmium pollution of soil-rice ecosystems in rice cultivation dominated regions in China: A review. *Environ. Pollut.* **2021**, *280*, 116965.
(20) Zheng, J.; Li, M.; Tang, B.; Luo, W.; Ma, Y.; Ren, M.; Yu, Y.; Luo, X.; Mai, B. Levels, Spatial Distribution, and Impact Factors of Heavy Metals in the Hair of Metropolitan Residents in China and Human Health Implications. *Environ. Sci. Technol.* **2021**, *55* (15), 10578−10588.
(21) Grau-Perez, M.; Voruganti, V. S.; Balakrishnan, P.; Haack, K.; Goessler, W.; Franceschini, N.; Redón, J.; Cole, S. A.; Navas-Acien, A.; Tellez-Plaza, M. Genetic variation and urine cadmium levels: ABCC1 effects in the Strong Heart Family Study. *Environ. Pollut.* **2021**, *276*, 116717.
(22) Tian, J.; Hu, J.; Chen, M.; Yin, H.; Miao, P.; Bai, P.; Yin, J. The use of mrp1-deficient (Danio rerio) zebrafish embryos to investigate the role of Mrp1 in the toxicity of cadmium chloride and benzo[a]pyrene. *Aquat. Toxicol.* **2017**, *186*, 123−133.
(23) Liu, Y.; Wang, X.; Si, B.; Wang, T.; Wu, Y.; Liu, Y.; Zhou, Y.; Tong, H.; Zheng, X.; Xu, A. Zinc oxide/graphene oxide nanocomposites efficiently inhibited cadmium-induced hepatotoxicity via releasing Zn ions and up-regulating MRP1 expression. *Environ. Int.* **2022**, *165*, 107327.
(24) Liu, J.; Bai, Y.; Feng, Y.; Liu, X.; Pang, B.; Zhang, S.; Jiang, M.; Chen, A.; Huang, H.; Chen, Y.; et al. ABCC1 deficiency potentiated noise-induced hearing loss in mice by impairing cochlear antioxidant capacity. *Redox Biol.* **2024**, *74*, 103218.
(25) Love, M. I.; Huber, W.; Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* **2014**, *15* (12), 550.
(26) Subramanian, A.; Tamayo, P.; Mootha, V. K.; Mukherjee, S.; Ebert, B. L.; Gillette, M. A.; Paulovich, A.; Pomeroy, S. L.; Golub, T. R.; Lander, E. S.; et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc. Natl. Acad. Sci. U. S. A.* **2005**, *102* (43), 15545−15550.
(27) Cole, S. P. C.; Deeley, R. G. Transport of glutathione and glutathione conjugates by MRP1. *Trends Pharmacol. Sci.* **2006**, *27* (8), 438−446.
(28) Wiese, M.; Stefan, S. M. The A-B-C of small-molecule ABC transport protein modulators: From inhibition to activation-a case study of multidrug resistance-associated protein 1 (ABCC1). *Med. Res. Rev.* **2019**, *39* (6), 2031−2081.
(29) Cosentino, A.; Agafonova, A.; Modafferi, S.; Trovato Salinaro, A.; Scuto, M.; Maiolino, L.; Fritsch, T.; Calabrese, E. J.; Lupo, G.; Anfuso, C. D.; et al. Blood-Labyrinth Barrier in Health and Diseases: Effect of Hormetic Nutrients. *Antioxid. Redox Signaling* **2024**, *40* (7−9), 542−563.
(30) Nyberg, S.; Abbott, N. J.; Shi, X.; Steyger, P. S.; Dabdoub, A. Delivery of therapeutics to the inner ear: The challenge of the blood-labyrinth barrier. *Sci. Transl. Med.* **2019**, *11* (482), No. eaao0935.
(31) Gu, J.; Tong, L.; Lin, X.; Chen, Y.; Wu, H.; Wang, X.; Yu, D. The disruption and hyperpermeability of blood-labyrinth barrier mediates cisplatin-induced ototoxicity. *Toxicol. Lett.* **2022**, *354*, 56−64.
(32) Sekulic-Jablanovic, M.; Paproth, J.; Sgambato, C.; Albano, G.; Fuster, D. G.; Bodmer, D.; Petkovic, V. Lack of NHE6 and Inhibition of NKCC1 Associated With Increased Permeability in Blood Labyrinth Barrier-Derived Endothelial Cell Layer. *Front. Cell. Neurosci.* **2022**, *16*, 862119.
(33) Chang, Y.; Jiang, X.; Dou, J.; Xie, R.; Zhao, W.; Cao, Y.; Gao, J.; Yao, F.; Wu, D.; Mei, H.; et al. Investigating the potential risk of cadmium exposure on seizure severity and anxiety-like behaviors through the ferroptosis pathway in epileptic mice: An integrated multi-omics approach. *J. Hazard. Mater.* **2024**, *480*, 135814.
(34) Deng, P.; Fan, T.; Gao, P.; Peng, Y.; Li, M.; Li, J.; Qin, M.; Hao, R.; Wang, L.; Li, M.; et al. SIRT5-Mediated Desuccinylation of RAB7A Protects Against Cadmium-Induced Alzheimer's Disease-Like Pathology by Restoring Autophagic Flux. *Adv. Sci.* **2024**, *11* (30), No. e2402030.
(35) Branca, J. J. V.; Morucci, G.; Pacini, A. Cadmium-induced neurotoxicity: still much ado. *Neural Regen. Res.* **2018**, *13* (11), 1879−1882.
(36) Sridhar, A.; DeSantiago, J.; Chen, H.; Pavel, M. A.; Ly, O.; Owais, A.; Barney, M.; Jousma, J.; Nukala, S. B.; Abdelhady, K.; et al. Modulation of NOX2 causes obesity-mediated atrial fibrillation. *J. Clin. Invest.* **2024**, *134* (18), No. e175447.
(37) Wang, Y.; Liu, X.-Y.; Wang, Y.; Zhao, W.-X.; Li, F.-D.; Guo, P.-R.; Fan, Q.; Wu, X.-F. NOX2 inhibition stabilizes vulnerable plaques by enhancing macrophage efferocytosis via MertK/PI3K/AKT pathway. *Redox Biol.* **2023**, *64*, 102763.
(38) Yingze, Y.; Zhihong, J.; Tong, J.; Yina, L.; Zhi, Z.; Xu, Z.; Xiaoxing, X.; Lijuan, G. NOX2-mediated reactive oxygen species are double-edged swords in focal cerebral ischemia in mice. *J. Neuroinflammation* **2022**, *19* (1), 184.
(39) Yi, S.-J.; Xiong, Y.-W.; Zhu, H.-L.; Dai, L.-M.; Cao, X.-L.; Liu, W.-B.; Shi, X.-T.; Zhou, G.-X.; Liu, A. Y.; Zhao, L.-L.; et al. Environmental cadmium exposure during pregnancy causes diabetes-like phenotypes in mouse offspring: Association with oxidative stress in the fetal liver. *Sci. Total Environ.* **2021**, *777*, 146006.
(40) Xu, C.; Wang, X.; Gu, C.; Zhang, H.; Zhang, R.; Dong, X.; Liu, C.; Hu, X.; Ji, X.; Huang, S.; et al. Celastrol ameliorates Cd-induced neuronal apoptosis by targeting NOX2-derived ROS-dependent PP5-JNK signaling pathway. *J. Neurochem.* **2017**, *141* (1), 48−62.
(41) Yao, L.; Liang, X.; Liu, Y.; Li, B.; Hong, M.; Wang, X.; Chen, B.; Liu, Z.; Wang, P. Non-steroidal mineralocorticoid receptor antagonist finerenone ameliorates mitochondrial dysfunction via PI3K/Akt/eNOS signaling pathway in diabetic tubulopathy. *Redox Biol.* **2023**, *68*, 102946.
(42) Deng, R.-M.; Zhou, J. The role of PI3K/AKT signaling pathway in myocardial ischemia-reperfusion injury. *Int. Immunopharmacol.* **2023**, *123*, 110714.
(43) Fang, S.; Wan, X.; Zou, X.; Sun, S.; Hao, X.; Liang, C.; Zhang, Z.; Zhang, F.; Sun, B.; Li, H.; et al. Arsenic trioxide induces macrophage autophagy and atheroprotection by regulating ROS-dependent TFEB nuclear translocation and AKT/mTOR pathway. *Cell Death Dis.* **2021**, *12* (1), 88.
(44) He, C.; Wang, K.; Xia, J.; Qian, D.; Guo, J.; Zhong, L.; Tang, D.; Chen, X.; Peng, W.; Chen, Y.; et al. Natural exosomes-like nanoparticles in mung bean sprouts possesses anti-diabetic effects via activation of PI3K/Akt/GLUT4/GSK-3β signaling pathway. *J. Nanobiotechnol.* **2023**, *21* (1), 349.
(45) Peng, J.; He, J.; Hu, X.; Xia, Y. GPR30 alleviated subarachnoid hemorrhage-induced blood-brain barrier dysfunction by activating the PI3K/Akt and Nrf2/HO-1 pathways. *Am. J. Physiol. Cell Physiol.* **2024**, *327* (1), C65−C73.
(46) Brasil, F. B.; Bertolini Gobbo, R. C.; Souza de Almeida, F. J.; Luckachaki, M. D.; Dall'Oglio, E. L.; de Oliveira, M. R. The signaling pathway PI3K/Akt/Nrf2/HO-1 plays a role in the mitochondrial protection promoted by astaxanthin in the SH-SY5Y cells exposed to hydrogen peroxide. *Neurochem. Int.* **2021**, *146*, 105024.

Downloaded from pubs.acs.org/ehnea2/article-pdf/doi/10.1021/envhealth.6c00120/6581560/envhealth.6c00120.pdf by guest on 10 October 2026

https://doi.org/10.1021/envhealth.6c00120
Environ. Health XXXX, XXX, XXX−XXX